



## End Term Examination May-June 2025

Roll Number.....

Name of the Program and Semester: B.Tech (CSE), IV Semester

Name of the Course: Fundamental of Statistics and AI

Course Code: TCS-421

Time: 3 Hours

Maximum Marks 100

### NOTE:

- (i) All the questions are compulsory.
- (ii) Answer any two sub-questions from (a), (b), and (c) in each main question.
- (iii) Total marks for each question are 20 (Twenty).
- (iv) Each sub-question carries 10 marks.

### Q1

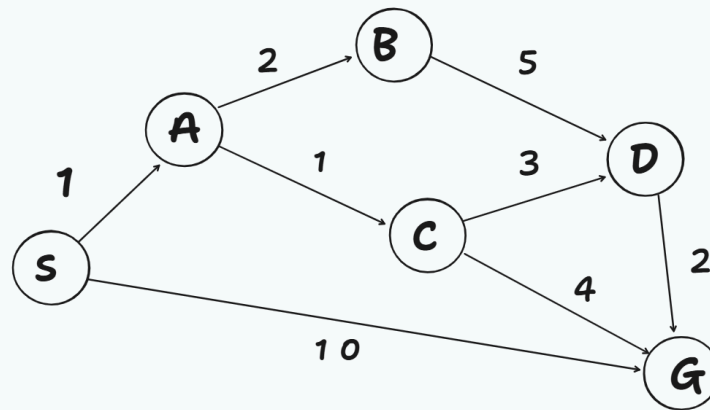
- a. Introduce Search techniques in AI that are Brute Force, Greedy and Heuristic.
- b. Define and give examples of Intelligent Agents in AI, give 2 applications for each Agent.
- c. Give Four different ways of defining AI. With block diagram outline typical AI Engine.

### Q2

- a. Explain and Describe the Types of AI in detail, giving an example for each type.
- b. Explain the functions in PEAS description, give real-world example that explains these functions.
- c. Apply A\* for the given graph to search the goal node. Consider Start Node = S and Goal Node = G, Table below gives the heuristic wrt goal node.

Heuristic Table:

| Node | S | A | B | C | D | G |
|------|---|---|---|---|---|---|
| h(n) | 6 | 7 | 8 | 4 | 6 | 0 |



**Q3**

- a. Discuss Turing Test and give two real-world applications.
- b. Explain big data using the 5 V's AND Outline the big data architecture with Hadoop.
- c. Explain what Data Science means AND describe the role of Data Scientist.

**Q4**

- a. Define the following with 1 example for each term:

1. Correlation
2. Random variable
3. Variance
4. Sample Mean

- b. Showcasing real-world example like online Product reviews for e-commerce providers like Amazon or Flipkart, justify Big data and back-end analytics. Also, explain the role of 5 V's in Big data.

- c. Illustrate the properties of Probability mass function, Cumulative Distribution Function and Probability density Function. Give two examples for each of these statistical functions.



## Q5

a. A fair coin is tossed 10 times, compute showing steps Probability:

1. At least 5 heads
2. At most 3 tails

State Bayes Theorem.

b. Describe the following:

1. Bernoulli Random variable
2. Outlier
3. A\* star heuristic
4. AI as Problem solving

c. Given the following data points depicting weight = W and length = L for Salmon obtain linear Regression model of L in terms of W.

|   |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|
| L | 10 | 12 | 16 | 18 | 19 | 21 | 25 |
| W | 22 | 44 | 49 | 52 | 59 | 42 | 51 |